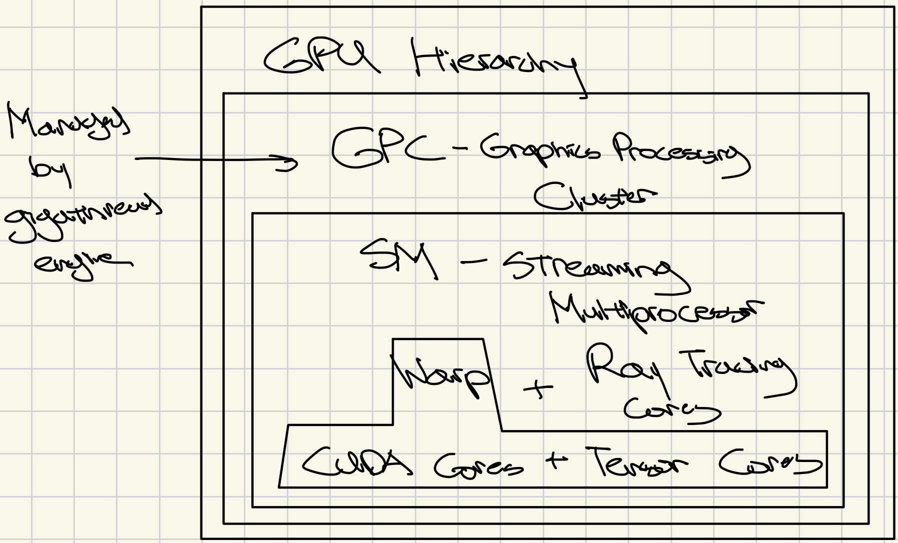
GPU Architecture

GPU	vs	CPU
- more cores w/ slower execution rate		- less cores w/ faster execution rate
- limited to certain tasks		- flexible, can run OS etc



Cores execute all calculations of the GPU

Cuda Cores - ALUs
- Mostly fused multiply & add $(A \times B + C)$
- Either 32 bit FP or INT
- One multiply + add per clock cycle
- A few special functions units per SM
  - Div / sqrt / trig

Tensor Cores
- Does $A \times B + C = D$ concurrently
  - Parallel mult
  - Addition with parallel reduction tree

Memory Architecture
- Data constantly streamed from SDRam to L2 cache on chip
  - Done by memory controllers
- Newest gen can encode in multiple voltages
  - Binary to Ternary - look more into
  - PAM 3/4

SIMD
- Single Instruction Multiple Data
- Same instruction on a lot of numbers
  - Ex. Transforming and moving object
    - Same instruction to move origin also used to move rest of object
  - Like one opcode used on multiple lanes with diff data

Thread Architecture
- 1 instruction per thread
- bundle of 32 threads → warp → all have same (SIMD) instruction
- warps grouped into thread blocks handled by SM
- Thread blocks grouped into grids

SIMT (newer architecture)
- Single instruction multiple threads
- Individual threads can progress at diff rates
- L1 cache shared by all threads in SM
- More flexibility dealing with warp divergence
  - If branching based on data
    - Still able to progress and synchronize instead of stalling on one warp